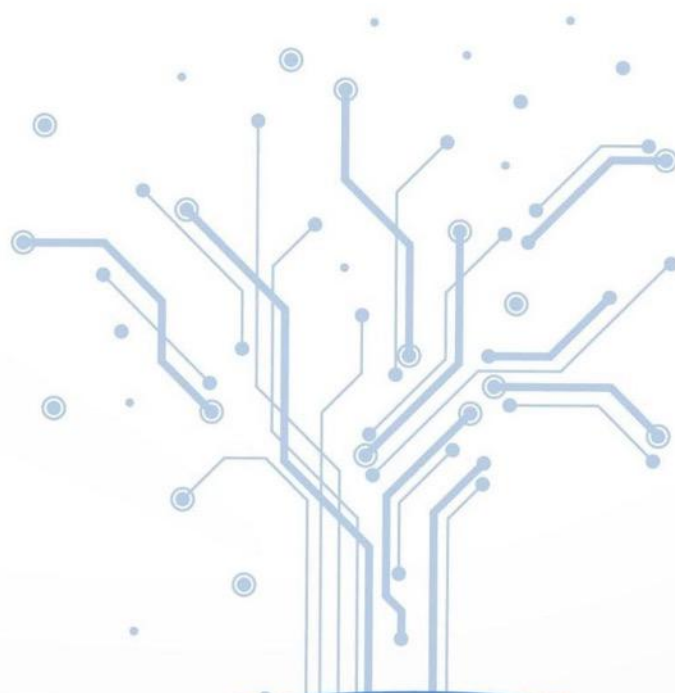




Animal visit reporting Hardware specifications



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Title: Democratising digital farming through tailored open source and open hardware solutions

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1 Hardware specifications

Hardware Title: UHF RFID smart case

Short Description: Smart device for controlling an RFID reader, pre-processing read data (aggregation per second), recording it and transmitting it to the cloud or to a mobile app via Bluetooth. The equipment is designed for continuous field collection of UHF RFID data for animals identified with UHF RFID tags. It allows monitoring of animal traffic in areas of interest. Watering, feeding. It can be used with all types of livestock (cattle, sheep, goats, pigs).

Keywords/Tags: UHF RFID,

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This device is designed in the form of a case containing a mini-PC (BeagleBoard), an RFID reader and a real-time clock. The device has indicator LEDs, USB charging ports, motion detector ports, and network ports for data communication. It is controlled and configured using a mobile application that sends commands to the mini PC via Bluetooth pairing.

Devices/ Equipment:

- [Beagleboard black with 512 MB RAM](#), ARM processor
- Bluetooth module
- Real Time Clock
- [RFID UHF Impinj reader R420](#) / R700
- [Mti antennas](#)
- [Smartrac UHF cattle tag](#)
- [Motion sensors](#)
- [Electric station, Ecoflow Delta 2 Max](#)
- [Solar Panel, EcoFlow 220W Bifacial](#)
- [4G/5G router - Huawei B535](#)
- [Linux server Debian 12.7](#)

Operating System: The OS (installed on the Beagleboard) has the following functions: It controls the reader, monitors the correct operation of the device, sends data to the DB and uses dedicated commands to communicate with a setting mobile app.

UHF ear tags to the reader: protocols communication is defined by the ISO 18000 pool of standards and the recently published ISO 6881 standard.

Networking: Device to database (cloud mode) with the use of 4G/5G connection

Networking: Device to farmer's phone (edge mode) with the use of standard Bluetooth

Power Source: Power Pack, electric station EcoFlow Delta 2 Max, and solar panel, EcoFlow 220W Bifacial, to bring electric power directly on-field and rechargeable.

Energy Consumption metering for the edge device: Current and voltage data logger use to measuring the consumption of complete RFID UHF device (RFID reader, 4G router, mini-pc). Read range: up to 6 meters
Power consumption: 6 Wh (stand-by mode), 18 Wh (UHF reading)

Licensing and Usage Rights: Chosen open hardware license (CERN OHL-S v2.0); Attribution required; Commercial use allowed under share-alike conditions.

Versioning and changelog: v3.5 – stable hardware release

